

Divisible design graphs derived from collections of affine designs

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Definition

A finite simple k -regular graph Γ on v vertices is a *divisible design graph* with parameters $(v, k, \lambda_1, \lambda_2, m, n)$ if the vertex set of Γ can be partitioned into m classes of size n such that

- any two different vertices from the same class have λ_1 common neighbors,
- any two vertices from different classes have λ_2 common neighbors.

If $m = 1$, $n = 1$, or $\lambda_1 = \lambda_2$, then a divisible design graph is a strongly regular with parameters (v, k, λ, μ) , where $k = \mu = \lambda_1$, $\lambda = \lambda_2$ or $\lambda = \mu = \lambda_1 = \lambda_2$.

The concept of divisible design graph was introduced by H. Kharaghani.

Divisible design graphs were first studied by W.H. Haemers, H. Kharaghani and M. Meulenberg in 2011.

In particular, the authors have proposed several constructions of divisible design graphs using various combinatorial structures.

[HKM2011] W.H. Haemers, H. Kharaghani, M. Meulenberg, *Divisible design graphs*, J. Combinatorial Theory, Series A, 118 (2011) 978–992.

Divisible design graphs is particularly interesting for its spectral properties. Just like strongly regular graphs, their eigenvalues can be determined using parameters [Lemma 2.1 HKM2011].

The spectrum of any divisible design graph with parameters $(v, k, \lambda_1, \lambda_2, m, n)$ can be calculated as follows:

$$\{k^1, \sqrt{k - \lambda_1}^{f_1}, -\sqrt{k - \lambda_1}^{f_2}, \sqrt{k^2 - \lambda_2 v}^{g_1}, -\sqrt{k^2 - \lambda_2 v}^{g_2}\}.$$

For the multiplicities of the eigenvalues, we generally have only the constraints

$$f_1 + f_2 = m(n - 1), \quad g_1 + g_2 = m - 1.$$

Using information about the eigenvalues of a divisible design graph, the authors of [HKM2011] obtain some necessary conditions for its existence.

Equitable partition of divisible design graphs

The partition from the definition of divisible design graphs is called the *canonical partition*.

Definition

A partition $\pi = \{C_1, \dots, C_m\}$ of the vertices of a graph is **equitable** if for every pair of (not necessarily distinct) indices $i, j \in \{1, \dots, m\}$ there is a nonnegative integer $c_{i,j}$ such that each vertex in C_i has exactly $c_{i,j}$ neighbors in C_j , regardless of the choice of the vertex.

Orbits of a group acting on a graph always form an equitable partition of the vertex set.

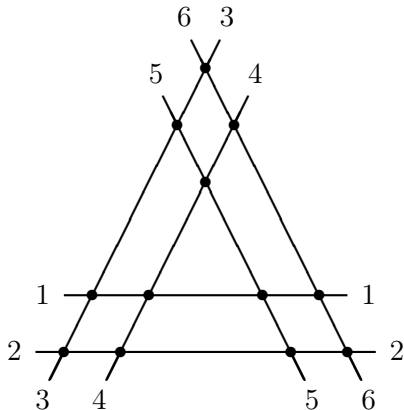
Theorem 3.1 [HKM2011]

The canonical partition of a divisible design graph is equitable.

The canonical partition matrix $C_\pi = (c_{i,j})$ of a divisible design graph is called the **quotient matrix**.

Small example

One of the interesting examples of a divisible design graph has parameters $(12, 6, 2, 3, 3, 4)$. This divisible design graph is known as the line graph of the octahedron.



In the figure, any two vertices on the same line are adjacent. Six lines represent 4-cliques and correspond to the six vertices of the octahedron.

Definition

Let G be a finite group with the identity element e . If S is a subset of G which is closed under inversion and does not contain e , then *Cayley graph*

$$\mathbb{C}ay(G, S)$$

is a graph with the vertex set G and two vertices x, y are adjacent if and only if $xy^{-1} \in S$.

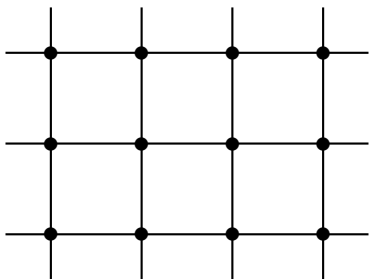
Let H be a proper subgroup of G .

The Cayley graph $\text{Cay}(G, G \setminus H)$ is a complete multipartite graph whose parts are cosets of H in G .

Obviously, a complete multipartite graph with equal parts is a divisible design graph.

Divisible design Cayley graphs

The $(4 \times n)$ -lattice is a divisible design Cayley graph on $C_4 \times C_n$ with parameters $(4n, n + 2, n - 2, 2; 4, n)$.



This picture depicted (4×3) -lattice whose parameters are $(12, 5, 1, 2, 4, 3)$.

The vertices on the vertical lines form canonical classes.

The following statement is an important property of divisible design Cayley graphs.

Proposition 1. [KSh2021]

If $\Gamma = \text{Cay}(G, S)$ is a divisible design Cayley graph, then the class of the canonical partition of Γ which contains the identity is a subgroup of G , and the canonical partition coincides with the right cosets partition of the group.

[KSh2021] V.V. Kabanov, L. Shalaginov, *On divisible design Cayley graphs*, Art of Discrete and Applied Mathematics, 4(2) (2021) #P2.02.

Definition

Let G be a finite group of order mn and N be a subgroup of G of order n . Then a subset S of G is called a **divisible difference set with the exceptional subgroup** N if there are constants λ_1 and λ_2 such that every non-identity element of N can be expressed as a right quotient of elements in S in exactly λ_1 ways and every element in $G \setminus N$ can be expressed as a right quotient of elements in S in exactly λ_2 ways.

From Proposition 1 it follows that a divisible design Cayley graph always arises through a divisible difference set with respect to some subgroup.

Construction of divisible design Cayley graphs

Let F be a finite field of order q^d , where q is a prime power, and $d > 1$.

It is known that the affine group $\text{Aff}(1, q^d)$ is the semidirect product of two subgroups N and K , where N is an elementary Abelian group of order q^d , and K is a cyclic group of order $q^d - 1$.

Let H be a subgroup of K of order $t = (q^d - 1)/(q - 1)$ generated by f .

Thus, $G = NH$ is a group of order $q^d(q^d - 1)/(q - 1)$.

Construction of divisible design Cayley graphs

Let $\mathbb{M}$ be a class of conjugate subgroups of order q^{d-1} of G which has exactly $m = (q^d - 1)/(q - 1)$ subgroups.

Let us fix a natural order on the set $\mathbb{M} = \{M_1, M_2, \dots, M_m\}$, defined the sequential conjugation by f for some M_1 .

Let φ be a permutation of the set $[m] := \{1, \dots, m\}$.

We define a subset S_φ of G as follows:

$$S_\varphi = \bigcup_{i=1}^m f^{\varphi(i)}(N \setminus M_i).$$

It is obvious that S_φ is a generating set of G .

The subset S_φ is closed under inversion in G if and only if the following condition holds for all i

$$\varphi^i(i) = m - i,$$

where $i \in [m]$.

For example, there are exactly three permutations for $m = 7$ that satisfy the above condition.

$(1, 6, 4, 2, 7, 5, 3)$, $(1, 6, 3, 7, 2, 4, 5)$, and $(1, 6, 2, 3, 5, 7, 4)$.

Construction of divisible design Cayley graphs

For any natural $m > 2$ there is at least one permutation φ satisfying the above condition.

If m is an odd integer, then

$$\varphi = (1, m-1, m-3, \dots, 2, m, m-2, m-4, \dots, 3).$$

If m is an even integer, then

$$\varphi = (1, m-1, m-3, \dots, m/2+2, m/2-1, m/2-3, \dots, \\ 2, m/2, m-2, \dots, m/2+1, m, m/2-2, m/2-4, \dots, 3).$$

Let

$$\Gamma_\varphi = \text{Cay}(G, S_\varphi),$$

where G is the normal subgroup of index $q - 1$ of the affine group $\text{Aff}(1, q^d)$ and

$$S_\varphi = \bigcup_{i=1}^m f^{\varphi(i)}(N \setminus M_i),$$

where φ is satisfied the condition $\varphi^i(i) = t - i$.

Construction of divisible design Cayley graphs

Theorem. [V.K. and L. Shalaginov 2021]

The graph Γ_φ is a divisible design graph with parameters

$$\begin{aligned}v &= q^d(q^d - 1)/(q - 1), & k &= q^{d-1}(q^d - 1), \\ \lambda_1 &= q^{d-1}(q^d - q^{d-1} - 1), & \lambda_2 &= q^{d-2}(q - 1)(q^d - 1), \\ m &= (q^d - 1)/(q - 1), & n &= q^d.\end{aligned}$$

Hypothesis. Γ_{φ_1} and Γ_{φ_2} are isomorphic if and only if

$$\langle \varphi_1 \rangle = \langle \varphi_2 \rangle.$$

[KSh2021] V.V. Kabanov, L. Shalaginov, *On divisible design Cayley graphs*, Art of Discrete and Applied Mathematics, 4(2) (2021) #P2.02.

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The smallest examples of divisible design Cayley graphs

There is exactly one divisible design Cayley graph with parameters $(12, 6, 2, 3, 3, 4)$ on $\text{Aff}(1, 4) \cong \text{Alt}_4$. This divisible design graph is the line graph of the octahedron.

There are exactly three non-isomorphic divisible Cayley graphs with parameters $(36, 24, 15, 16, 4, 9)$, two of which are derived from the construction.

There are exactly five non-isomorphic divisible design Cayley graphs with parameters $(56, 28, 12, 14, 7, 8)$, three of which are derived from the construction.

Definition

A k -regular graph Γ on v vertices is called a *strongly regular graph with parameters* (v, k, λ, μ) if any two adjacent vertices have exactly λ common neighbors, while any two nonadjacent vertices have exactly μ common neighbors.

Strongly regular graphs with $\lambda = \mu$ constitute a prominent subclass of the family of divisible design graphs.

Various methods for construction of strongly regular graphs have been developed. A fruitful approach to constructing such graphs involves the use of affine designs. In 1971, Wallis proposed a new construction of strongly regular graphs using a collection of affine designs organized by a Steiner 2–design.

Fon-Der-Flaass later found a way to modify a specific case of the Wallis construction, when the corresponding Steiner 2–design has blocks of size 2, in order to obtain hyperexponentially many strongly regular graphs with the same parameters.

Subsequently, Muzychuk demonstrated how to generalize Fon-Der-Flaass's approach to all instances of the Wallis construction, thereby identifying several new families of strongly regular graphs.

[WW1971] W.D. Wallis, *Construction of strongly regular graphs using affine designs*, Bull. Austral. Math. Soc., 4 (1971), 41–49.

[FF2002] D.G. Fon-Der-Flaass *New prolific constructions of strongly regular graphs*, Adv. Geom., 2 (2002), 301–306.

[MM2007] M. Muzychuk, *A generalization of Wallis-Fon-Der-Flaass construction of strongly regular graphs*, J. Algebr. Comb., 25 (2007) 169–187.

Definition

An *affine design* $\mathcal{D}$ with parameters p and q is a design, where the set of points and the set of blocks have the following two properties:

- (i) every two blocks are either disjoint or intersect in p points;
- (ii) each block together with all blocks disjoint from it forms a parallel class: a set of q mutually disjoint blocks partitioning all points of the design.

If $\mathcal{D}$ is an affine design with parameters p and q , then all parameters of D are expressed in terms of p and q :

q^2p is the number of points,
 qp is the block size,
 $(q^2p - 1)/(q - 1)$ is the number of parallel classes.

Any d -dimensional affine space over a finite field of order q is the point-hyperplane design with parameters q, q^{d-2} . This design has q^d points and $(q^d - 1)/(q - 1)$ parallel classes.

Other known examples of affine designs are finite Hadamard 3-designs, Desarguesian and non-Desarguesian finite planes.

In [VK2022] and [VK2026], we introduced several constructions of divisible design graphs using a collection of affine designs organized by the Cayley table of a left quasigroup. These constructions develop the ideas of Wallis, Fon-Der-Flaass, and Muzychuk in the context of divisible design graphs.

[VK2022] V.V. Kabanov, *New versions of the Wallis–Fon-Der-Flaass construction to create divisible design graphs*, Discrete Mathematics, 345(11) (2022) Article ID 113054.

[VK2026] V.V. Kabanov, *Construction of divisible design graphs using affine designs*, Disc. Math., 349(2) (2026) Article ID 114717.

Some of divisible design graphs from [VK2022] were used in [VK2023] to construct new strongly regular graphs with parameters of the complement of the symplectic graph. This was made possible by decomposing of a strongly regular graph into a divisible design graph and a Hoffman coclique.

[VK2023] V.V. Kabanov, *A New Construction of Strongly Regular Graphs with Parameters of the Complement Symplectic Graph*,

Possible parameters of strongly regular graphs decomposable into a divisible design graph and either a Hoffman coclique or a Delsarte clique were studied in [GK2023] and [GK2024].

Electron. J. Combin., 30(1) (2023) # P1.25,

[GK2024] A.L. Gavriluk, V.V. Kabanov, *Strongly regular graphs decomposable into a divisible design graph and a Hoffman coclique*. Des. Codes Cryptogr., 92 (2024) 1379–1391.

[GK2025] A.L. Gavriluk, V.V. Kabanov, *Strongly regular graphs decomposable into a divisible design graph and a Delsarte clique*. Des. Codes Cryptogr., 93, (2025) 2177–2189.

Combinatorial construction of divisible design graphs

In this talk, we propose a new approach to constructing divisible design graphs using a collection of affine designs.

- To construct a new divisible design graph, we select a collection of affine designs with identical parameters.
- The union of their point sets is used as a set of graph vertices.
- Indexing the parallel classes of each design and using them to create ‘special table’.
- Defining arbitrary bijections between block sets based on a ‘special table’ of their indexes.
- Adjacency is established by relating each block in one affine design to its image (or its complement) in the other design according to the given bijection.

This approach yields hyperexponentially many divisible design graphs relative to the number of their vertices.

Let V be a d -dimensional affine space over $\mathbb{F}_q$ and $\mathcal{D} = \{\mathcal{D}_i = (\mathcal{P}_i, \mathcal{B}_i) \mid i \in [\hat{m}]\}$ be a collection of copies of point-hyperplane affine designs over V .¹

Let us fix a $(d-s)$ -dimensional subspace W of V , where $0 \leq d-s \leq d-1$. For each $\mathcal{D}_i$, we enumerate the parallel classes of blocks generated by those hyperplanes of V that do not contain W . The number of such classes is given by:

$$\hat{m} = \frac{q^d - 1}{q - 1} - \frac{q^s - 1}{q - 1} = q^s \frac{q^{d-s} - 1}{q - 1}.$$

¹Throughout this text, $[n]$ denotes the set $\{1, \dots, n\}$ for any $n \in \mathbb{N}$.

Let $L = (e(i, j))_{1 \leq i, j \leq \hat{m}}$ be a matrix such that the entries of each row represent a random permutation of integers from $[\hat{m}]$. Thus, L can be interpreted as the Cayley table of a left quasigroup on $[\hat{m}]$.

For each pair (i, j) , we choose an arbitrary bijection

$$\sigma_{i,j} : \mathcal{B}_i^{e(i,j)} \rightarrow \mathcal{B}_j^{e(j,i)}.$$

subject to the condition $\sigma_{i,j} = \sigma_{j,i}^{-1}$. Furthermore, we let $\sigma_{i,i}$ be the identity bijection on $\mathcal{B}_i^{e(i,i)}$. Let Σ denote the set of all such bijections.

Let $\Gamma = \Gamma(\mathcal{D}, L, \Sigma)$ be a graph defined as follows:

- The vertex set of Γ is $V(\Gamma) = \bigcup_{i=1}^{\hat{m}} \mathcal{P}_i$.
- Two distinct vertices $x \in \mathcal{P}_i$, and $y \in \mathcal{P}_j$ are adjacent in Γ if and only if

$$y \notin \sigma_{i,j}(B_i^{e(i,j)}(x)) \quad \text{for } i, j \in [\hat{m}],$$

where $B_i^{e(i,j)}(x)$ denotes the block in the $e(i,j)$ -th parallel class of $\mathcal{D}_i$ containing x .

Theorem 1. [V.K. 2026]

Let $\Gamma = \Gamma(\mathcal{D}, L, \Sigma)$ be a graph obtained from Construction 1. Then Γ is a divisible design graph with parameters:

$$v = \frac{q^{d+s}(q^{d-s} - 1)}{q - 1}, \quad k = q^{d+s-1}(q^{d-s} - 1),$$

$$\lambda_1 = q^{d+s-1}(q^{d-s} - q^{d-s-1} - 1),$$

$$\lambda_2 = q^{d+s-2}(q^{d-s} - 1)(q - 1),$$

$$m = \frac{q^{2s}(q^{d-s} - 1)}{q - 1}, \quad n = q^{d-s}.$$

The spectrum of Γ has four distinct eigenvalues

$$\{q^{d+s-1}(q^{d-s} - 1), \quad q^{d-1}, \quad 0, \quad -q^{d-1}\}.$$

The multiplicities of the non-principal eigenvalues are

$$\frac{q^s(q^{d-s} - 1)(q^d - q^s - q + 1)}{2(q - 1)},$$

$$\frac{q^{d+s} - q^{2s} - q + 1}{q - 1},$$

$$\frac{q^s(q^{d-s} - 1)(q^d - q^s + q - 1)}{2(q - 1)}.$$

Let V be a d -dimensional affine space over a finite field $\mathbb{F}_q$, where q is a power of 2. Let $\mathcal{D} = \{\mathcal{D}_i = (\mathcal{P}_i, \mathcal{B}_i) \mid i \in [\hat{m}]\}$ be a collection of copies of point-hyperplane affine designs over V . The same as in previous section let W be a fixed $(d - s)$ -dimensional subspace of V , where $0 \leq d - s \leq d - 1$. For each $\mathcal{D}_i$, we enumerate the parallel classes of blocks generated by those hyperplanes of V that do not contain W . The number of such classes is given by:

$$\hat{m} = q^s \frac{q^{d-s} - 1}{q - 1}.$$

Let $L = (e(i, j))$ be an $(\hat{m} \times \hat{m})$ -matrix acting as the Cayley table of a left quasigroup on $[\hat{m}]$.

For each pair (i, j) , choose an arbitrary bijection

$$\sigma_{i,j} : \mathcal{B}_i^{e(i,j)} \rightarrow \mathcal{B}_j^{e(j,i)},$$

subject to the condition $\sigma_{i,j} = \sigma_{j,i}^{-1}$.

In this construction, we further require that $\sigma_{i,i}$ has no fixed blocks on $\mathcal{B}_i^{e(i,i)}$. This distinguishes the current approach from Construction 1, where the mappings $\sigma_{i,i}$ were identities.

Let Σ denote the set of these bijections.

Let $\Gamma = \Gamma(\mathcal{D}, L, \Sigma)$ be a graph defined as follows:

- The vertex set of Γ is $V(\Gamma) = \bigcup_{i=1}^{\hat{m}} \mathcal{P}_i$.
- Two distinct vertices $x \in \mathcal{P}_i$ and $y \in \mathcal{P}_j$ are adjacent in Γ if and only if

$$y \in \sigma_{i,j}(B_i^{e(i,j)}(x)) \quad \text{for } i, j \in [\hat{m}].$$

Even though $y \in \sigma_{i,j}(B_i^{e(i,j)}(x))$ is opposite to $y \notin \sigma_{i,j}(B_i^{e(i,j)}(x))$ in the previous construction, this does not lead to the complement of a graph since we chose another bijections on $\mathcal{B}_i^{e(i,i)}$.

Theorem 2. [V.K. 2026]

Let $\Gamma = \Gamma(\mathcal{D}, L, \Sigma)$ be a graph obtained from Construction 2. Then Γ is a divisible design graph with parameters:

$$v = \frac{q^{d+s}(q^{d-s} - 1)}{q - 1}, \quad k = \frac{q^{d+s-1}(q^{d-s} - 1)}{q - 1},$$

$$\lambda_1 = \frac{q^{d+s-1}(q^{d-s-1} - 1)}{q - 1},$$

$$\lambda_2 = \frac{q^{d+s-2}(q^{d-s} - 1)}{q - 1},$$

$$m = \frac{q^{2s}(q^{d-s} - 1)}{q - 1}, \quad n = q^{d-s}.$$

Constructions using point-hyperplane designs

The spectrum of Γ is determined by the parameters. It turns out that Γ has four distinct eigenvalues

$$\left\{ \frac{q^{d+s-1}(q^{d-s} - 1)}{q - 1}, \quad q^{d-1}, \quad 0, \quad -q^{d-1} \right\}.$$

The multiplicities of the non-principal eigenvalues are

$$\begin{aligned} & \frac{q^s(q^{d-s} - 1)(q^d - q^s - 1)}{2(q - 1)}, \\ & \frac{q^{d+s} - q^{2s} - q + 1}{q - 1}, \\ & \frac{q^s(q^{d-s} - 1)(q^d - q^s + 1)}{2(q - 1)}. \end{aligned}$$

- When $s = 1$ in Construction 1 as well as Construction 2, we can use a set of arbitrary affine designs with the same parameters instead of point-hyperplane designs, and fix one parallel class in each $\mathcal{D}_i$ instead of choosing a $(d - 1)$ -dimensional subspace W of V .
- Since $q^d - q^{d-1} = q^{d-1}$, for $q = 2$, a divisible design graph in Theorem 2 with $q = 2$ has the same parameters as in Theorem 1.

A divisible design graph from Theorem 2 with $q = 2$ and $s = d - 1$ has parameters

$$(2^{2d-1}, \quad 2^{2d-2}, \quad 0, \quad 2^{2d-3}, \quad 2^{2d-2}, \quad 2).$$

Crnković and Haemers called a divisible design graph *thin* if $n = 2$. This is exactly what we have in this case.

An $(1, -1, 0)$ -matrix W is called a weighing matrix of order n and weight k if

$$WW^T = kI_n,$$

where I_n is the identity matrix of order n and W^T denotes the transpose of W . A weighing matrix of order n and weight n is called a Hadamard matrix.

It was demonstrated in [Section 4 CH2014] that a thin divisible design graph admits the quotient matrix R and a corresponding weighing matrix Q of the same size such that if

$$A_{ij} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$$

is a canonical block of the adjacency matrix, then the corresponding elements of Q and R are defined by

$$(Q)_{i,j} = a - b, \quad (R)_{i,j} = a + b,$$

respectively.

This means that from the adjacency matrix of any thin divisible design graph from Construction 2 we can obtain a symmetric Hadamard matrix of order 2^{2d-2} , having a constant main diagonal.

[CH2014] D. Crnković and W. H. Haemers, *Walk-regular divisible design graphs*, Des. Codes Cryptogr. **72** (2014), 165–175.

Partial case of Construction 1

Theorem 1 from [VK2022] is partial case of Theorem 1 when $s = 0$.

In this case Γ has parameters

$$v = q^d(q^d - 1)/(q - 1), \quad k = q^{d-1}(q^d - 1),$$

$$\lambda_1 = q^{d-1}(q^d - q^{d-1} - 1), \quad \lambda_2 = q^{d-2}(q - 1)(q^d - 1),$$

$$m = (q^d - 1)/(q - 1), \quad n = q^d.$$

and four distinct eigenvalues $\{q^{d-1}(q^d - 1), q^{d-1}, 0, -q^{d-1}\}$.

[VK2022] V.V. Kabanov, New versions of the Wallis-Fon-Der-Flaass construction to create divisible design graphs, *Discrete Mathematics*, 345(11) (2022) Article ID 113054.

This case is applicable to constructing strongly regular graphs.

Cocliques in strongly regular graphs

Let Γ be a strongly regular graph with parameters (v, k, λ, μ) , and eigenvalues k, r, s , where $r > 0 > s$.

The coclique number in strongly regular graphs satisfies the well-known Delsarte-Hoffman bound.

If C is a coclique in Γ , then $|C| \leq v/(1 + \frac{k}{-s})$.

If the equality holds, then C is called a *Hoffman coclique*.

If a strongly regular graph has a Hoffman coclique C , then there is a special case of regular decompositions and the induced subgraph on $V(\Gamma) \setminus C$ has at most four eigenvalues.

W.H. Haemers and D.G. Higman studied the case where the induced subgraph on $V(\Gamma) \setminus C$ has three distinct eigenvalues, which means that a strongly regular graph admits a decomposition into a strongly regular graph and a Hoffman coclique.

[HH] W.H. Haemers and D.G. Higman, Strongly Regular Graphs with Strongly Regular Decomposition, Linear Algebra Appl., 114–115, (1989) 379–398.

Construction of strongly regular graphs with Hoffman coclique

Strongly regular graphs that can be decomposed into a divisible design graph and a Hoffman coclique were found in [VK2023].

Namely, a prolific construction of strongly regular graphs with parameters of the complement of the symplectic graph were provided by using divisible design graphs from the partial case of Construction 1.

[VK1] V.V. Kabanov, A new construction of strongly regular graphs with parameters of the complement symplectic graph, Electron. J. Comb., 30(1) (2023) # P1.25

All the found examples of strongly regular graphs have the smallest eigenvalue, which is a prime power.

The smallest possible example whose smallest eigenvalue is -6 is a strongly regular graph with parameters $(259, 216, 180, 180)$. This graph is unknown.

A divisible graph with necessary parameters $(252, 210, 174, 175, 7, 36)$ is also unknown.

This brings us to the problem of 36 officers.

If there was an affine plane with 36 points, we could use our construction to find such graph.

However, if there is a necessary divisible design graph, then there is a strongly regular graph as well.

Thank you for your attantion